

ETL hop nhap du lieu giao thong Non-IID tai 3 node bien

Nguồn dữ liệu: tọa độ thu công trong config/nodes.yaml, TomTom Traffic Flow, OSM Overpass, ảnh 3 điểm.

How to transform: chuẩn hóa schema, tính congestion ratio, LOS, density proxy, gan lineage, hop nhap co trong so.

Kết quả hop nhap node

node_id	node_name	status	fused_velocity	fused_congestion_ratio	los	congestion_level	confidence	observation_count	weight_sur
N01_LY_THUONG_KIET	Ly Thuong Kiet	ok	32.7778	0.0	E	thoang	1.0	9	1.0
N02_BA_THANG_HAI	Ba Thang Hai	ok	28.7972	0.0252	E	trung_binh	0.996	9	1.0
N03_CMT8	Cach Mang Thang Tam	ok	34.0	0.0	E	thoang	1.0	9	1.0
N04_THANH_THAI	Thanh Thai	ok	30.7778	0.0	E	thoang	1.0	9	1.0
N05_TO_HIEN_THANH	To Hien Thanh	ok	30.2222	0.0	E	thoang	1.0	9	1.0
N06_NGUYEN_TRI_PHUONG	Nguyen Tri Phuong	ok	33.3333	0.0	E	thoang	1.0	9	1.0
N07_SU_VAN_HANH	Su Van Hanh	ok	28.6667	0.0	E	trung_binh	1.0	9	1.0
N08_DIEN_BIEN_PHU	Dien Bien Phu	ok	41.0	0.0	F	thoang	1.0	9	1.0

Thống kê Non-IID

node_id	count	mean_velocity	std_velocity	min_velocity	max_velocity	skew_velocity
N01_LY_THUONG_KIET	9	32.7778	2.4845	30.0	35.0	-0.2236
N02_BA_THANG_HAI	9	28.7778	5.0062	20.0	35.0	-0.267
N03_CMT8	9	34.0	0.0	34.0	34.0	0.0
N04_THANH_THAI	9	30.7778	1.0304	29.0	32.0	-0.7649
N05_TO_HIEN_THANH	9	30.2222	0.4157	30.0	31.0	1.3363
N06_NGUYEN_TRI_PHUONG	9	33.3333	1.4907	32.0	35.0	0.2236
N07_SU_VAN_HANH	9	28.6667	3.8006	22.0	33.0	-0.9081
N08_DIEN_BIEN_PHU	9	41.0	0.0	41.0	41.0	0.0

Kiểm định phân phối

node_a	node_b	ks_distance	js_distance	interpretation
N01_LY_THUONG_KIET	N02_BA_THANG_HAI	0.5556	0.7588	different_distribution
N01_LY_THUONG_KIET	N03_CMT8	0.5556	1.0	different_distribution
N01_LY_THUONG_KIET	N04_THANH_THAI	0.5556	1.0	different_distribution
N01_LY_THUONG_KIET	N05_TO_HIEN_THANH	0.5556	0.6497	different_distribution
N01_LY_THUONG_KIET	N06_NGUYEN_TRI_PHUONG	0.4444	0.7103	different_distribution
N01_LY_THUONG_KIET	N07_SU_VAN_HANH	0.5556	0.7854	different_distribution
N01_LY_THUONG_KIET	N08_DIEN_BIEN_PHU	1.0	1.0	different_distribution
N01_LY_THUONG_KIET	N09_CONG_HOA	0.8889	1.0	different_distribution

Trạng thái dữ liệu: Live TomTom data collected from Traffic Flow API.

Cửa sổ thu thập: ['00:00-23:59']; GitHub auto collection: 6 tháng.